The examination is being carried out on the **following application documents**

Description, Pages

1-32, 34-96	as published	
33	filed in electronic form on	15-10-2025

Claims, Numbers

1-15	filed in electronic form on	15-10-2025

Drawings, Sheets

1/88-24/88, 26/88-88/88	as published	
25/88	filed in electronic form on	15-10-2025

1 The following documents (D) are referred to in this communication; the numbering will be adhered to in the rest of the procedure:

D1 US 2021/092018 A1 (FANG YU [US] ET AL) 25 March 2021 (2021-03-25)

D2 US 2016/255154 A1 (KIM DONG KYUE [KR] ET AL) 1 September 2016 (2016-09-01)

2 The application does not meet the requirements of Article 84 EPC in that the matter of independent **claims 1 and 9** and dependent **claims 10-13** for which protection is sought is not clearly defined for the following reasons:

2.1 The subject-matter for which protection is sought by independent **claim 1** is unclear because no interworking relation is defined between, on the one hand, the feature according to which the plurality of network zones is arranged according to a network topology description of a selected zone architecture, said selected zone architecture being determined in response to a vehicle network risk description and a vehicle performance network impact description, and, on the other hand, the features of the network management controller relating to the zoned architecture communication scheme and the zone

manager command descriptions. According to the Guidelines for Examination, Part F-IV, 4.1, 4.2 and 4.5, the meaning of the terms of a claim and the technical relationships between its essential features should be clear to the person skilled in the art from the wording of the claim alone. However, claim 1 does not define that the zoned architecture communication scheme is established in accordance with the selected zone architecture and then e.g. forwarded and processed as input of the zone implementation circuit (for receiving the zoned architecture communication scheme as input see paragraph [80]: "...wherein interpreting the zoned architecture communication scheme includes receiving a policy from an external device at the central zone manager..."), that the zone manager command descriptions take account of the network topology description, or that the selected zone architecture otherwise influence the operation of the network management controller.

It is therefore not possible to understand from the wording of claim 1 whether the risk- and performance-based selection of the zone architecture has any technical effect on the generation or execution of the zone manager command descriptions. The architecture-selection features and the communication-control features consequently appear as isolated groups of features whose technical interrelation and combined technical effect are undefined, thereby rendering the definition of the subject-matter of said claim 1 unclear by the description within the meaning of Article 84 EPC.

2.2 The subject-matter of independent **claim 1** is furthermore unclear because the function of the “zone manager command circuit” and the origin of the “zone manager command description” are not unambiguously defined. Claim 1 specifies that the zone manager command circuit is structured to “interpret a zone manager command description for each of the at least one zone managers in response to the zoned architecture communication scheme”. This wording indicates that the zone manager command description already exists as an input to the zone manager command circuit. At the same time, the expression “in response to the zoned architecture communication scheme” could be understood as meaning that the zone manager command circuit generates or determines the zone manager command description from the communication scheme. The description, in particular paragraphs [401, 402], discloses that the zone manager command circuit parses a policy or configuration file in order to determine the zone manager command descriptions, which may then be communicated to the respective zone managers as local configuration files. Additionally the description, in particular paragraph [396], discloses that "the zone manager commend circuit 5006 determines the responsibilities of each zone manager to be responsive to the zoned architecture communication

scheme 5014, and determines the zone manager command descriptions 5016 to implement the portions of the zoned architecture communication scheme 5014 that are relevant to that particular zone manager". The wording of claim 1 therefore does not clearly reflect the disclosed input/output relationship. It is consequently unclear whether the zone manager command description is an input which is merely read or interpreted by the zone manager command circuit or an output which is generated or determined by that circuit. It is also unclear which specific information from the zoned architecture communication scheme is converted into the command description associated with each respective zone manager. The applicant is invited to clarify the functional chain, for example by defining that the zone manager command circuit determines, from the zoned architecture communication scheme, a respective zone manager command description for each zone manager. Moreover, the expressions “for each of the at least one zone managers” and “each of the at least one zone managers” are grammatically inconsistent, since “at least one” is singular whereas “zone managers” is plural. The subject-matter of claim 1 is therefore additionally unclear contrary to Article 84 EPC.

2.3 The subject-matter for which protection is sought by independent **claim 9** is unclear because the features "vehicle network risk description", "vehicle performance network impact description", "zonal architecture value" and "integrating a multi-zone network" are not defined with regard to their technical content, in particular the “zonal architecture value”. Additionally the input/output relationships of the "vehicle network risk description” and the “vehicle performance network impact description” is undefined. In particular, claim 9 merely specifies that a zonal architecture value is determined in response to a vehicle network risk description and a vehicle performance network impact description. The claim does not define a proposed or candidate zonal architecture, the technical parameters contained in the zonal architecture value, the manner in which the performance network impact description is determined, or the evaluation by which the architecture value is selected or modified. According to paragraphs [410-412] of the description, the vehicle performance network impact description may be determined in response to the network routing layout, network topology, positions of network components, the zoned architecture communication scheme and information concerning the vehicle and its operating environment. The zonal architecture value, as the "initial zonal architecture value is utilized as a starting point" may subsequently be adjusted on the basis of performance and cost parameters until an acceptable value or a convergence criterion is reached. These essential technical relationships are absent from claim 9.

Furthermore, the step of “integrating a multi-zone network on a vehicle in response to the zonal architecture value” defines only a result to be achieved and does not specify which physical or configurational steps are performed to reach this result. The description refers more specifically to installing the network in accordance with the zonal architecture value, its network topology description or its zone manager distribution description. Claim 9 does not define any of these technical implementation measures. According to the Guidelines for Examination, Part F-IV, 4.5 and 4.10, an independent claim must contain the essential features necessary to achieve the claimed technical result, and a definition merely by a result to be achieved is normally not allowable where the essential technical means for achieving that result are omitted. Since claim 9 neither defines the technical content of its principal input and output data nor the processing steps linking those data to the installed multi-zone network, the subject-matter of claim 9 is not clearly defined and is not adequately supported by the description, contrary to Article 84 EPC.

2.4 The subject-matter for which protection is sought by dependent **claims 13** is unclear because firstly the same objections apply to "vehicle network performance description" as described in paragraph 2.3 with regard to the "vehicle performance network impact description". Additionally "determining a zonal architecture value in response to the vehicle network performance description and the vehicle performance network impact description" in claim 13 contradicts "determining a zonal architecture value in response to the vehicle network risk description and a vehicle performance network impact description". Claim 13 depends on claim 9 and it is thus not clear whether the "zonal architecture value" is determined "in response to the vehicle network performance description and the vehicle performance network impact description" or "in response to the vehicle network risk description and a vehicle performance network impact description". Therefore the subject-matter of claim 13 is not clearly defined contrary to Article 84 EPC.

2.5 The subject-matter of dependent **claims 10-12** is unclear because the expressions "hardware redundancy value", "control redundancy value" and "critical flow description" do not define objective technical limits or are relative terms. **Claims 10 and 11** do not define whether the respective redundancy “value” identifies components for which redundancy is required, a number of redundant components, a required redundancy level, an acceptable degradation of operation following a failure, or another technical parameter. The description indicates that these values may identify hardware or control elements for which redundancy is to be provided and may additionally specify acceptable performance levels during redundant operation. These limitations

are not present in the claims. **Claim 12** likewise does not define the technical criteria according to which a flow is to be regarded as "critical". The term "critical" is a relative term whose boundary may depend on the intended vehicle function, operating condition, failure consequence or required redundancy level. Although the description provides examples of critical flows and applications and refers to acceptable performance in the event of their loss, the claim neither specifies these criteria nor defines how the identified critical flow affects the topology, zone-manager distribution or another parameter of the determined zonal architecture value. According to the Guidelines for Examination, Part F-IV, 4.6, a relative term is potentially unclear unless it has a well-recognised meaning in the relevant technical field or its meaning and limits are clear in the context of the disclosure. Claims 10-12 should therefore be amended to define the respective redundancy requirement and the technical criterion for criticality, together with the manner in which that requirement or criterion is taken into account when determining the zonal architecture value. Until such clarification is provided, said claims do not meet the requirements of Article 84 EPC.

3 It is noted that if the applicant does not succeed in removing all raised objections the application is likely to be refused under Article 97(2) EPC. It is further noted that any suggestions are only for assisting the applicant in his decision on how to proceed. It in no way precludes consideration of alternative solutions submitted by the applicant. The responsibility for determining the text of the application (Article 113(2) EPC) and in particular for defining the subject-matter for which protection is sought remains with the applicant.

4 When amending the claims, the applicant should also take the opportunity to attend to the following points:

4.1 In order to meet the requirements of Rule 42(1)(b) EPC the relevant prior art, i.e. the documents D1-D2, noted above, should be acknowledged by reference and briefly discussed in the introductory part of the description, preferably in such a way that the inventive merit of what is claimed can be readily understood. D1 discloses a vehicle comprising a plurality of network zones and a distributed converged network device having portions arranged to regulate communications between endpoints of different network zones. D1 further discloses network zones having different risk-exposure profiles and selected topologies including bus, serial, mesh, hub, ring and star topologies (see in particular paragraphs [209-223]). D1 additionally discloses, in particular in paragraphs [302-320], interpreting a policy comprising a network-regulation

description, configuring network-interface circuits in response to that policy and operating the interface circuits to regulate communications between vehicle networks. The policy manager may update configuration files and translate high-level policy descriptions into executable commands for network-regulating components. D1 is therefore particularly relevant to the features of claim 1 concerning a zoned communication policy, commands or configurations derived from that policy, distributed network-regulating components and the control of communications between endpoints of different zones.

D2 discloses a method and apparatus for designing a vehicular security network by assessing risk levels of vehicle functional elements, grouping those functional elements into security zones based on the assessed risk levels and selecting and implementing gatekeepers or security countermeasures appropriate to the respective zones. D2 further teaches that a network-connectivity topology based on vehicle structure, roles and types of functional elements, design requirements and system performance may be taken into account when setting the security zones (see in particular paragraphs [14-18] and [84-89]). D2 is therefore particularly relevant to the features of claim 9 relating to the interpretation of vehicle-network risk information, the determination of a zonal arrangement in view of risk, topology and system-performance considerations and the subsequent implementation of zone-related security components. Notwithstanding the clarity objections raised in paragraphs 2.1 to 2.5 above, neither D1 nor D2 appears to disclose all features of the independent claims 1 and 9.

It is nevertheless noted that D1 and D2 relate to closely associated aspects of vehicle network zoning. Once the subject-matter of claims 1 and 9 has been clearly defined, it will have to be examined whether the person skilled in the art, starting from D1 and seeking to select its network-zone arrangement in view of vehicle risk and performance considerations, would take account of the risk- and performance-based zone-setting teaching of D2. In particular, the applicant may be required to demonstrate a technical interaction extending beyond the mere juxtaposition of the risk- and performance-based architecture-selection features known from D2 and the policy-based inter-zone communication-control features known from D1.

4.2 All the claims should include reference signs in parentheses where features shown in the drawings are referred to (Rule 43(7) EPC).

4.3 The opening part of the description should be brought into conformity with the wording of the claim of broadest scope as finally amended.

4.4 The "incorporated by reference" statement in paragraphs [4, 273, 369] should be deleted, in accordance with the EPO-Examination-Guidelines, F-III-8.

4.5 The applicant should correct the following typing mistake in paragraph [396]: "zone manager command circuit 5006" instead of "zone manager commend circuit 5006".

4.6 The Applicant should adapt the description to the claims on file and delete or amend any statements or expressions which throw doubt on the scope of protection (GL F-IV, 4.3). Failure to do so could lead to the refusal of the application under Article 97(2) EPC in combination with Article 84 EPC. The applicant should amend the description as needed to fulfil the following requirements: The technical field of the invention (Rule 42(1)(a), GL F-II.4.2) must correspond to the amended set of claims. The summary of the invention (Rule 42(1)(c) and GL F-II.4.5) must correspond to the amended set of claims. If appropriate, statements such as “The invention is set out in the appended set of claims“ or "The invention is as defined in claim X" can be applied instead of repeating the claims verbatim. All embodiments which are not covered by the claims anymore must be deleted, unless these embodiments can reasonably be considered to be useful for highlighting specific aspects of the amended subject-matter. In such case, they must be prominently marked as not being part of the present invention (T 1808/06). Subject-matter in the description being excluded from patentability under Article 53 EPC needs to be excised, reworded such that it does not fall under the exceptions to patentability or prominently marked as not being according to the claimed invention. All technical features of the independent claims shall not be presented as optional in the description. Terms such as “for example“, “may“, “can”, “exemplary“, “optionally“, “preferably“ and the like should be removed when preceding a feature of an independent claim. Claim-like clauses must be deleted prior to grant. The term “claim-like clauses“ means clauses present in the description which use claim language such as “according to the preceding clause“, “characterised in that“, … (GL F-IV, 4.4). These “claim-like clauses“ also include examples, embodiments or aspects presented as claim-like clauses. A simple conversion of these clauses into separate optional features will not be admitted, either.

4.7 Special care should be taken to avoid giving rise to further objections by the inadvertent addition of subject-matter (Article 123(2) EPC). In this respect it is to be noted that the indication, in the letter of reply, of the part(s) of the originally filed application serving as basis for subject-matter newly introduced into the

claims by the applicant would be useful since it would greatly facilitate the subsequent substantive examination of the claims with respect to Article 123(2) EPC.

4.8 Finally, amendments to the application should be filed by way of typewritten or printed replacement pages (Rule 49(8) EPC), but only for amended pages. Handwritten submissions and amendments to the description, claims and abstract – unless they involve graphic symbols and characters and chemical and mathematical formulae – are a formal deficiency under Rules 49(8) and 50(1) EPC, possibly in conjunction with Rule 86 EPC.